

Advanced Fast Breeder Thorium Reactor (AFBTR)

Concept Proposal

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Executive Summary

The Advanced Fast Breeder Thorium Reactor (AFBTR) is a conceptual Generation-IV-inspired nuclear reactor designed to support long-term sustainable energy production using India's abundant thorium resources. The proposed reactor combines fast breeder reactor technology, thorium fuel breeding, passive safety systems, and advanced waste reduction strategies to address major challenges associated with conventional nuclear power.

Motivation

India possesses approximately 25% of the world's known thorium reserves, while domestic uranium resources remain comparatively limited. Current nuclear technologies rely heavily on uranium-based fuel cycles. The AFBTR concept aims to utilize thorium as a long-term energy resource while improving fuel efficiency, reducing waste generation, and enhancing reactor safety.

Core Design Features

Reactor Type: Fast Breeder Reactor (FBR)

Fuel Cycle: Startup Fuel: Uranium-233 or Plutonium-239; Blanket Material: Thorium-232; Coolant: Liquid Sodium

Electrical Output: ~500 MW

Breeding Objective: Breeding Ratio > 1.0

Thorium Breeding Mechanism

$\text{Th-232} + n \rightarrow \text{Th-233} \rightarrow \text{Pa-233} \rightarrow \text{U-233}$. The resulting Uranium-233 can be recycled as reactor fuel, enabling a sustainable fuel cycle.

Passive Safety Systems

Gravity-Driven Emergency Cooling: Automatic activation, natural circulation, no external power required.

Molten Salt Backup Drain Tank: Passive shutdown via freeze plug melting and safe fuel drainage.

Waste Reduction Strategy

Reprocessing of spent fuel, use of MOX fuel, and consumption of minor actinides reduce long-lived radioactive waste.

Potential Benefits for India

Utilization of domestic thorium resources, reduced dependence on uranium imports, improved energy security, and lower carbon emissions.

Current Limitations

Requires advanced fuel reprocessing, detailed simulations, economic feasibility studies, and experimental validation.

Future Research Directions

Focus on reactor simulations, breeding optimization, neutron economy, and safety modeling.

Author

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